

Group B

Attempt any SIX questions

2. Why is the study of errors important to a computational scientist? Differentiate between inherent and numerical errors.
3. Find the root of equation $x^2 - 4x - 10 = 0$ using bisection method where root lies between 5 and 6.
4. Find the square root of 3.5 using second order Lagrange interpolation polynomial using the following data table.

x	1	2	3	4	5
f(x)	1	1.4142	1.7321	2	2.2361

5. Write a program to calculate the integral using Trapezoidal Rule.
6. Solve the following set of equations using Gauss-Jordan Method.
 $3x - 5y + 2z = 15$
 $4x - y + z = 2$
 $x - 3y + 7z = 22$
7. Use classical Runge-Kutta method to estimate $y(0.2)$ when $y'(x) = x^2 + y^2$ with $y(0) = 0$ and $h = 0.2$.
8. Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f=0$ on the boundary and $h = 1$.

Group C

Attempt any TWO questions.

9. a) Factorize the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$ Using Cholesky's method.
- b) Locate the root of equation $x^2 + x - 2$ using the fixed point method.
10. a) Fit a straight line to the following set of data points.

x	1	3	4	6	8	9
y	1	3	4	4	5	7

- b) Differentiate between ordinary and partial differential equations with their applications.
11. Given the data points:

i	0	1	2
x_i	2	3	4
f_i	4	9	16

Estimate the function value f at $x = 2.5$ using cubic splines.

2019 Batch

Group B

Attempt any SIX questions

- If the true value of π is 3.1415926 and its approximate value is given by 3.1428571. Find the absolute and relative errors.
- What are the advantages of Lagrange's formula over Newton's formula? When Newton's forward interpolation formula is used?
- Find the polynomial that interpolates $f(0) = 0$, $f(1) = 1$, $f(2) = 0$, $f(3) = 1$, $f(4) = 0$ using Newton divided differences. Use the Newton table to generate the necessary divided differences.
- Using the trapezium rule, evaluate the integral $I = \int_0^1 \frac{dx}{x^2 + 6x + 10}$ with 2 and 4 sub intervals. Compare with the exact solution. Comment on the magnitudes of the errors obtained.
- For solving a linear system, compare Gauss elimination method and Gauss Jordan method. Why Gauss seidel method is better than Jacobi's iterative method?
- Using Euler's method find $y(0.2)$ from $\frac{dy}{dx} = x + y$, $y(0) = 1$ with $h = 0.2$.
- Solve the Poisson equation $\Delta^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

Group C

Attempt any TWO questions.

- When would we not use N-R method? Find the root of the equation $x^3 - 4x + 1 = 0$, lying in $(0, 1)$ Using Bisection method performing 10 iterations.
- Find the dominant eigen value of $A = \begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$ by power method. Apply Gauss seidel method to solve system of equations.
 $6x_1 - 2x_2 + x_3 = 11$; $-2x_1 + 7x_2 + 2x_3 = 5$; $x_1 + 2x_2 - 5x_3 = -1$, with the initial vector of $(0, 0, 0)$.
- Prepare the multi-step methods available for solving ordinary differential equation, Evaluate the value of y at $x=0.1$ and 0.2 to 4 decimal places given $\frac{dy}{dx} = x^2y - 1$, $y(0)=1$, using Taylor series method.

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2018 Batch

Group B

Attempt any SIX questions

- Compute the root of equation $x^2 - 5x + 6 = 0$ in the vicinity of $x = 5$ using Newton-Raphson method.
- Estimate the value of $\ln(2.5)$ using Newton-Gregory forward difference formula given the following data.

x	1.0	2.0	3.0	4.0
Ln(x)	0.0	0.6931	1.0986	1.3863

- Write an algorithm to compute integration using Simpson's 1/3 rule. Evaluate integration of $\int_0^2 (3x^2 + 2x - 5)dx$ using Simpson's 1/3 rule. [2.5 + 2.5]
What is meant by ill-conditioned system? Find the Cholesky decomposition of the following matrix. [5]

$$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

- Use the classical RK method to estimate $y(0.4)$ of the equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$ assume $h = 0.2$. [5]
- Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$. [5]
- Write a short note on (Any Two): [2.5 + 2.5]
 - Types of Errors
 - Convergent Methods
 - PDE

Group C

Attempt any TWO questions.

- Write an algorithm and program to compute root of nonlinear equation using bisection method. [2×10 = 20]
- Given then data points: [4 + 6]

i	0	1	2
x_i	9	16	25
f_i	3	4	5

Estimate the function value of f at $x = 14$ using cubic splines. [10]

- Solve the following system of equation by Gauss-Seidel method. [5]

$$\begin{aligned} 2x - 7y - 10z &= -17 \\ 5x + y + 3z &= 14 \\ X + 10y + 9z &= 7 \end{aligned}$$
 - Write an expansion of Taylor's Theorem. Solve the equation $y'(x) = x^2 + y^2$, $y(0) = 1$ for $x = 0.5$ using Taylor method. [2.5 + 2.5]

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2021 Batch

Group B

Attempt any SIX questions

- Explain absolute and relative error. Find the relative error of number 5.6 if both of its digits are correct.
- On what type of equations Newton's method can be applicable-Justify.
- Solve the following equations by using Gauss-Jordan method.
 $2x+3y+4z=5$ $3x+4y+5z=6$ $4x+5y+6z=7$
- Use the Romberg method to get an improved estimate of the integral from $x = 1.8$ to $x = 3.4$ from the data in the table with $h = 0.4$.

X	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8
Y	4.95	6.05	7.38	9.02	11.02	13.46	16.44	20.05	24.53	29.96	36.59	44.70
	3	0	9	5	3	4	5	6	3	4	8	1

- Write a program to compute integral $\int_0^{\pi/2} \sqrt{\sin x} dx$ Simpson's 1/3 rule.
- Using Runge-Kutta method of 4th order solve the following equation taking each step $h = 0.1$
 $\frac{dy}{dx} = \frac{4x}{y} - xy$ given $y(0) = 3$ calculate y at $x = 0.1$ and 0.2 .
- Solve the Laplace equation $U_{xx} + U_{yy} = 0$ for the following square mesh with the boundary values as shown in the figure below:



Group C

Attempt any TWO questions.

- Solve the given set of linear equations using Doolittle LU decomposition method:
 $3x_1 + 2x_2 + x_3 = 10$
 $2x_1 + 3x_2 + 2x_3 = 14$
 $3x_1 + 2x_2 + 3x_3 = 14$
- Define initial value problems and final value problems. Using Heun's method, find value of y when $x=0.3$ given that $\frac{dy}{dx} = x + y$ and $y=1$ when $x=0$.
- How can we use Interpolation techniques (methods) to approximate the value of the integral for the functions whose antiderivative can't be found? Explain. Write a program to solve $\sin x - 2x + 1 = 0$ using Bisection method.

2020 Batch

Group B

Attempt any SIX questions

- Define error. Explain the Taxonomy Errors.
- Estimate the value of $\sin \theta$ at $\theta = 45^\circ$ using Newton's backward difference formula from the following set of data.

θ	10	20	30	40	50	60
$\sin \theta$	0.1736	0.3420	0.5000	0.6428	0.7660	0.8660

- Write an algorithm and program to calculate integration using Trapezoidal rule.
- What is the form of resultant matrix using Gauss-Jordan method? Solve the following system of equations using Gauss-Jordan Method.]
 $x + 2y - 3z = 4$
 $2x + 4y - 6z = 8$
 $x - 2y + 5z = 4$
- Define ordinary differential equation. Use the fourth order Runge-Kutta method to estimate $y(0.4)$ of the equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$ assuming that $h = 0.2$.
- Solve for the steady-state temperatures in a rectangular plate of 8cm x 10cm, if one 10cm side is held at 50°C , and the other 10cm side is held at 30°C and other two sides are held at 10°C . Assume grids of size 2cm x 2cm.
- Write a short note on (Any Two):
a) Partial Differential Equations a) Linear Interpretation b) Boundary value problems

Group C

Attempt any TWO questions.

- Write an algorithm and program to compute the root of nonlinear equation using Newton - Raphson method.
- a) Fit a straight line to the following set of data using Least Square Regression.
b) Apply the factorization method (any) to solve the equations:
 $3x + 2y + 7z = 4$
 $2x + 3y + z = 5$
 $3x + 4y + z = 7$
- Write and implement an algorithm to solve the system of linear equations using Gauss-Seidel method with suitable numerical example.
b) Write a program to solve the ordinary differential equations using Heun's method.

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